

Andrew Vu

Cybersecurity Engineer



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SUMMARY

Computer Science graduate (Oregon State University, Cum Laude) with an undergraduate Cybersecurity Certificate and a security-first approach to software and systems engineering. Strong foundation in secure software development, access control, applied cryptography, network security, digital forensics, vulnerability assessment, and information security governance through cybersecurity coursework. Former NCAA Division I tennis player.

EDUCATION

Oregon State University — Corvallis, OR 09/2022 – 06/2026

B.S. in Computer Science | Cybersecurity Certificate (2026) | GPA: 3.51/4.0 | Cum Laude | Dean's List

Security Coursework: Network Security, Applied Cryptography, Digital Forensics, Security Governance, Usability Engineering

Core CS Coursework: Data Structures & Algorithms, Analysis of Algorithms, Operating Systems, Computer Networks, Database Systems, Parallel Programming

EXPERIENCE

StrokeVision — Co-Founder & Lead Engineer (Early-Stage Startup) 01/2026 – Present

React Native (Expo), TypeScript, Python, FastAPI, PostgreSQL, AWS S3, Secure API Handling

- Designed the app's security model: scoped each user's uploaded videos to that user in cloud storage (AWS S3), validated and constrained actions before execution, and flagged low-confidence inputs instead of guessing.
- Built the iOS app (React Native/Expo) on a Python/FastAPI backend with secure API handling and input validation across the client-server boundary; async pose-estimation pipeline runs in a background worker pool. TestFlight beta in progress.

Software Engineer (Freelance) — Gresham United Methodist Church, Gresham, OR 09/2025 – 12/2025

- Led a full platform migration (Wix to Framer) including domain transfer, DNS configuration, and email continuity to minimize downtime and avoid data loss during cutover.
- Integrated secure PayPal payment workflows and applied separation-of-concerns and input-handling practices while translating stakeholder requirements into controlled, maintainable changes.

Software Engineer Intern — Wholistic Peace Institute, Lake Oswego, OR 06/2025 – 08/2025

- Sole developer who designed, built, and deployed the organization's website end to end on Bluehost, including server configuration, hardening, and reliability.
- Built an API-driven checkout system on the Google Books API with secure PayPal/Stripe payment integrations and input validation.

PROJECTS

AI-Powered Home Assistant Integration — Senior Capstone | Python, OpenAI API, REST APIs, AsyncIO

- Built an AI Home Assistant integration with a secure execution layer that validates commands before they run, using an asynchronous architecture for concurrent request handling.

SponsorHub — Sponsorship Management Platform | Next.js, TypeScript, PostgreSQL, Prisma

- Built a full-stack platform with role-based access control (RBAC) enforcing authentication and authorization across user roles; modeled relational data in PostgreSQL via Prisma with background job processing (BullMQ, Redis).

TECHNICAL SKILLS

Core Security Competencies: Network Security, DNS, Threat Modeling, Vulnerability Assessment, Digital Forensics & Incident Response, Security Monitoring, Event Log Analysis, Applied Cryptography, Access Control & Authentication, IAM, RBAC, HMAC Authentication, Secure API Handling, Input Validation, Information Security Governance, Security Policy Development, Risk Assessment & Compliance, NIST Cybersecurity Framework, ISO 27001, ISO 27001 Concepts, Wireshark, PowerShell, Linux, SDLC, Agile

Languages: Python, Java, JavaScript, TypeScript, C/C++, SQL

Web & APIs: Node.js, React, Angular, Next.js, HTML5, CSS3, REST APIs, JSON, XML, JSON Schema, Postman, Swagger, API Testing

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, CI/CD, Git, GitHub, GitHub Actions, Linux, Windows

Databases: PostgreSQL, MySQL, MongoDB

AI/ML & Data: Artificial Intelligence (AI), Machine Learning, Neural Networks, NLP, LLMs, Data Mining, Data Analytics, Computer Vision

Methodologies: SDLC, Agile, Scrum, QA Testing

LEADERSHIP & ATHLETICS

- NCAA Division I Men's Tennis, Portland State University — multiple USTA Junior Championship titles. Student Ambassador, Oregon State University.